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$$f(x, y) = \sin(xy^2)$$

Eerste orde:

$$\frac{\partial f}{\partial x} = \cos(xy^2) \cdot y^2$$

$$\frac{\partial f}{\partial y} = \cos(xy^2) \cdot 2xy$$

Tweede orde:

$$\frac{\partial^2 f}{\partial x^2} = -\sin(xy^2) \cdot y^4$$

$$\frac{\partial^2 f}{\partial y^2} = -\sin(xy^2) \cdot 4x^2y^2 + \cos(xy^2) \cdot 2x$$

$$\frac{\partial^2 f}{\partial x \partial y} = -\sin(xy^2) \cdot 2xy^3 + \cos(xy^2) \cdot 2y$$

$$\frac{\partial^2 f}{\partial y \partial x} = -\sin(xy^2) \cdot 2xy^3 + \cos(xy^2) \cdot 2y$$

References